

A small 3D U-Net for lung tumor segmentation with anatomy-level 3D reconstruction: a reproducible CPU baseline with false-success controls

Lungs project contributors · technical report · 2026-09-25

Abstract

Purpose. We describe an open, fully reproducible pipeline that segments lung tumors on chest CT with a deliberately small 3D U-Net and turns each scan into an anatomy-level 3D model (five lobes, airway tree with centreline, pulmonary arteries and veins, and tumor). We report results with checks that catch *false success*. The whole thing runs on a 4-core CPU. **Methods.** We trained a 3.2M-parameter 3D U-Net on 45 patients from the Medical Segmentation Decathlon (MSD) Task06 Lung dataset. Scans were resampled to 1.5 mm isotropic and cropped to the lungs. We validated on 5 patients and tested on 13 held-out patients. Split is by patient. Lobes, airways and vessels come from the public TotalSegmentator models. We add a pruned airway centreline, branch-generation counting and patient-space meshes. Test metrics are computed at native resolution with bootstrap confidence intervals. We compare against an intensity-threshold baseline, an empty predictor and a shuffled-label control, and against the ceiling set by our own resampling. **Results.** On the test split the model reached tumor Dice 0.610 [0.450, 0.736], HD95 76.9 [39.6, 119.2] mm and lesion-level sensitivity 0.79 with 1.92 false-positive components per scan. It detected the tumor in 11 of 13 patients. A largest-component rule chosen on validation data gave Dice 0.658 [0.485, 0.787] with 0.08 FP/scan. The threshold baseline scored Dice 0.041 [0.000, 0.113]; the shuffled-label control scored 0.000 [0.000, 0.000]; the resampling ceiling was 0.946 [0.935, 0.957]. **Conclusion.** A small model trained in about 3.1 hours on a 4-core CPU (including validation) gives a clear, honestly bounded baseline. For context, nnU-Net reports roughly 0.7 Dice on the official MSD lung test set; that set differs from our internal split, so the numbers are not directly comparable. We release code, tests, the 3D viewer and every number in this report as machine-generated files. This is a research prototype and not a diagnostic device.

1 Introduction

Lung cancer is the leading cause of cancer death worldwide. CT is the main imaging test for finding and staging it. Automatic tumor segmentation supports volumetry, follow-up and treatment planning. A 3D model that shows the tumor next to the lobes, airways and vessels helps with surgical and bronchoscopic planning. Strong segmentation systems exist, such as nnU-Net [2] and TotalSegmentator [5]. But reported numbers are often hard to reproduce, and small studies can overstate success through patient leakage, lenient metrics or missing baselines.

This report makes three contributions. (i) A small, documented 3D U-Net baseline for MSD Task06 that can be trained end to end on a CPU. (ii) An anatomy-level reconstruction pipeline that places the predicted tumor in the context of lobes, airway tree (with centreline and endpoints) and vessels, and exports meshes in patient coordinates. (iii) An evaluation protocol whose negative controls and automated tests are built to expose false success, with every reported number generated from code.

2 Related work

The U-Net [1] and its 3D extension [3] are the standard architectures for volumetric medical segmentation. nnU-Net [2] showed that careful, automatically configured training matters more than architectural changes, and its Decathlon results [4] remain the reference point for this task, with lung tumor Dice of roughly 0.7. TotalSegmentator [5] offers robust open models for more than 100 structures, including lobes, airways and pulmonary vessels. LIDC-IDRI/LUNA16 [6] is the main benchmark for nodule detection. Metric pitfalls such as Dice on empty masks, unbounded boundary distances and resampling bias are catalogued in [7], which shaped our protocol.

3 Methods

3.1 Data

MSD Task06 Lung [4] contains 63 labelled thin-slice CT scans of patients with non-small-cell lung cancer (CC-BY-SA 4.0), with expert tumor masks. The 32 official test scans have no public labels, so we split the 63 labelled scans by patient with a fixed seed (42): 45 training, 5 validation and 13 test. Tests assert that no patient appears in more than one

split. All scans are LAS-oriented. We convert them to RAS for training and map predictions back to each original grid for evaluation.

3.2 Preprocessing

We clip HU to $[-1000, 400]$ and scale to $[0, 1]$. A classical lung mask is derived: threshold at -320 HU, remove air connected to the image border, keep the two largest components and fill holes. It is used to crop each scan ($+10$ mm margin) and as the lung class. The tumor label overrides it. Crops are resampled to 1.5 mm isotropic, trilinear for images and nearest for labels. The resulting three-class problem is background, lung and tumor.

3.3 Network and training

The 3D U-Net has four resolution levels, base width 12 (widths 12–192) and 3,176,799 parameters. Each level has two $3\times 3\times 3$ convolutions with InstanceNorm and LeakyReLU(0.01). Pooling is isotropic; upsampling uses transposed convolutions; the bottleneck has dropout 0.1. The loss is soft Dice over foreground classes (batch-level) plus cross-entropy with class weights (0.5, 1, 3). We train with AdamW (lr 0.003, wd 10^{-4}), a one-cycle schedule, gradient clipping at 12 and batch 2. Patches are $64\times 96\times 96$ voxels, 8 per patient per epoch, 66% centred on tumor voxels. Augmentations are left–right flip, gamma, contrast and Gaussian noise. We train for 20 epochs (about 9 min per epoch on 4 CPU cores) and keep the checkpoint with the best validation mean Dice (lung and tumor), checked every 4 epochs.

3.4 Inference

Inference uses a sliding window (patch $96\times 128\times 128$, 50% overlap) with Gaussian importance weighting and flip test-time augmentation. Post-processing keeps the two largest lung components and removes tumor components under 20 mm^3 or outside the closed lung hull. Tumor probabilities are mapped back to the native grid by trilinear interpolation and thresholded at 0.5.

3.5 Post-processing selection

The default rule keeps every tumor component $\geq 20\text{ mm}^3$ inside the lung hull. Because the first test run showed many false-positive components, we compared 18 alternative rules (minimum component volume, minimum mean probability, largest component only). We compared them on the *validation* patients only (lung_041, lung_045, lung_034, lung_048, lung_033). The best rule, `{'min_tumor_mm3': 20, 'min_mean_prob': 0.0, 'largest_only': True}`, raised validation Dice from 0.723 to 0.731 and cut FP/scan from 2.8 to 0.0. It was then applied unchanged to the test set. We report both rules. Because the aggregate test FP rate motivated this experiment, the second row of Table 1 is not a fully blind result, and with five validation patients the selection is itself noisy.

3.6 Anatomy-level reconstruction

Per patient, the five lobes and trachea come from TotalSegmentator's fast model, and the airway tree, pulmonary arteries and pulmonary veins from its `lung_vessels` model. We run the latter on a lung-cropped CT (about 44% of the voxels) to stay within 15 GB of RAM, then paste the result back. The airway is skeletonised in 3D, and terminal spurs shorter than 5 mm are pruned. The resulting graph gives branch points, terminal endpoints, centreline length and a Weibel-style generation count from the most superior tracheal point. Surfaces are extracted with marching cubes after Gaussian smoothing in physical units (2 mm for lobes, 0.5–0.8 mm for airway, vessels and tumor) and Taubin $\lambda|\mu$ smoothing [8], then mapped to RAS millimetres. Each tumor is assigned to the lobe it overlaps most. Meshes are exported as OBJ (full resolution, for 3D Slicer) and as a decimated web bundle for our viewer.

3.7 Evaluation and false-success controls

All test metrics are computed on the *native* CT grid against the original expert label. We report Dice, HD95 (mm, using the physical voxel spacing), voxel sensitivity and precision, lesion-level sensitivity and false-positive components per scan. Components count if they are ≥ 3 mm in equivalent diameter, on both sides. The expert labels contain 1–10-voxel specks that are not lesions, and counting them had artificially deflated lesion sensitivity in our first evaluation run. A lesion counts as detected only if the prediction covers $>10\%$ of it. Our first run used "any overlap", which counted a 1% touch of a 151 mL tumor as a hit. We also report case-level detection ($>10\%$ of the tumor covered) and volume error. 95% confidence intervals come from 2000 bootstrap resamples over patients. Controls: (a) an *empty* predictor; (b) an

HU-threshold baseline (soft tissue, -300 to 200 HU, inside the closed lung hull but outside aerated lung, largest component); (c) a *shuffled-label* control, scoring our prediction for patient i against the label of patient $i+1$; and (d) the *resampling ceiling*, the expert label sent through our 1.5 mm pipeline and back. The automated test suite (33 passed, 0 failed, 0 skipped) checks, among other things: empty-vs-non-empty masks give Dice 0 and infinite HD95, not NaN or 0; tumor-free cases do not inflate tumor Dice; an all-foreground predictor is penalised; an untrained network scores near zero; splits are disjoint and deterministic; sliding-window inference matches whole-volume inference exactly for a voxel-wise model; nodule volumes are physically correct under anisotropic spacing; and reconstructed tumor centroids match label centroids in RAS within 3 mm.

4 Results

Table 1. Tumor segmentation on the held-out MSD Task06 test split ($n = 13$ patients), evaluated at native CT resolution. Mean with bootstrap 95% CI over patients. HD95 in mm. Lesion sensitivity and FP/scan count connected components of at least 3 mm equivalent diameter on both sides (smaller expert-label specks are annotation noise). A lesion (or a case) counts as detected only if the prediction covers $>10\%$ of its volume.

Method	Dice	HD95 (mm)	Sensitivity	Precision	Lesion sens.	FP / scan	Case det.
Small 3D U-Net (ours, default post-proc.)	0.610 [0.450, 0.736]	76.9 [39.6, 119.2]	0.695 [0.505, 0.851]	0.609 [0.481, 0.709]	0.79	1.92	0.85
+ largest-component rule (val-selected)	0.658 [0.485, 0.787]	15.2 [4.2, 29.4]	0.691 [0.504, 0.846]	0.731 [0.587, 0.845]	0.79	0.08	0.85
HU-threshold baseline	0.041 [0.000, 0.113]	113.7 [86.9, 138.5]	0.025 [0.000, 0.069]	0.152 [0.000, 0.381]	0.07	0.85	0.08
Empty prediction (control)	0.000 [0.000, 0.000]	∞	0.000 [0.000, 0.000]	n/a	0.00	0.00	0.00

The small 3D U-Net clearly beats the trivial threshold rule and every control (Tables 1–2). The shuffled-label control stays near zero, so the score reflects patient-specific localisation rather than generic tumor-like blobs. Mean Dice was 0.65 for tumors ≥ 5 mL ($n = 8$) and 0.55 for tumors < 5 mL ($n = 5$); with groups this small the difference is not reliable. Predicted and expert tumor volumes correlated with Pearson $r = 0.39$, with mean absolute error 16.9 [1.8, 41.3] mL (Fig. 2).

Table 2. False-success controls. A trustworthy result must sit well above the controls and below the ceiling.

Check	Tumor Dice	Interpretation
Ours vs. expert label (default)	0.610 [0.450, 0.736]	main result
Ours vs. expert label (largest component)	0.658 [0.485, 0.787]	validation-selected rule
Ours vs. <i>another patient's</i> label	0.000 [0.000, 0.000]	should be ~ 0 ; high values would mean the metric rewards generic blobs
Expert label $\rightarrow$ 1.5 mm grid $\rightarrow$ native	0.946 [0.935, 0.957]	upper bound imposed by our resampling
HU-threshold baseline	0.041 [0.000, 0.113]	what a trivial intensity rule achieves
Empty prediction	0.000 [0.000, 0.000]	must be exactly 0

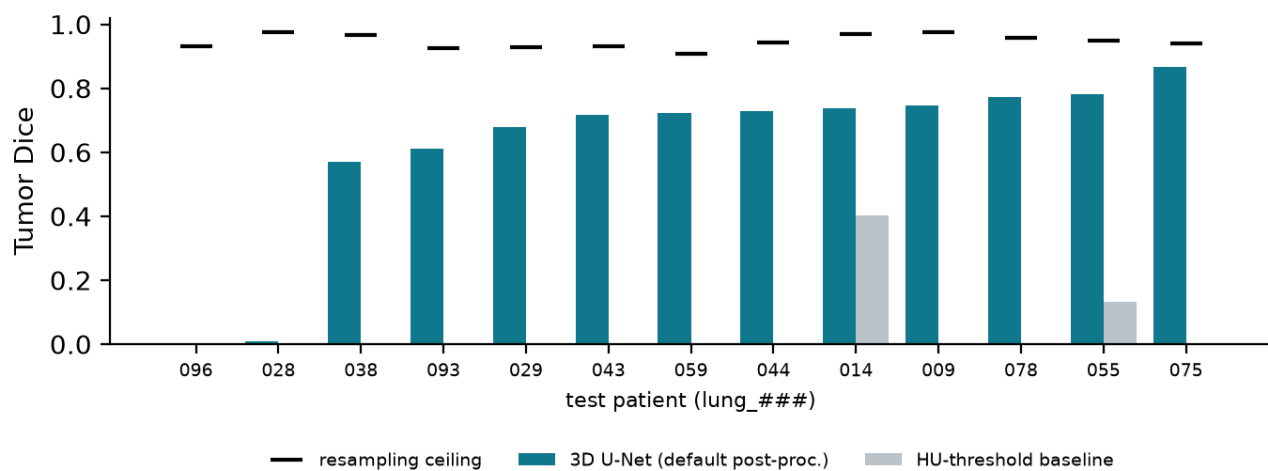


Figure 1. Per-patient tumor Dice on the test split for our model (default post-processing) and the threshold baseline; black ticks mark the resampling ceiling.

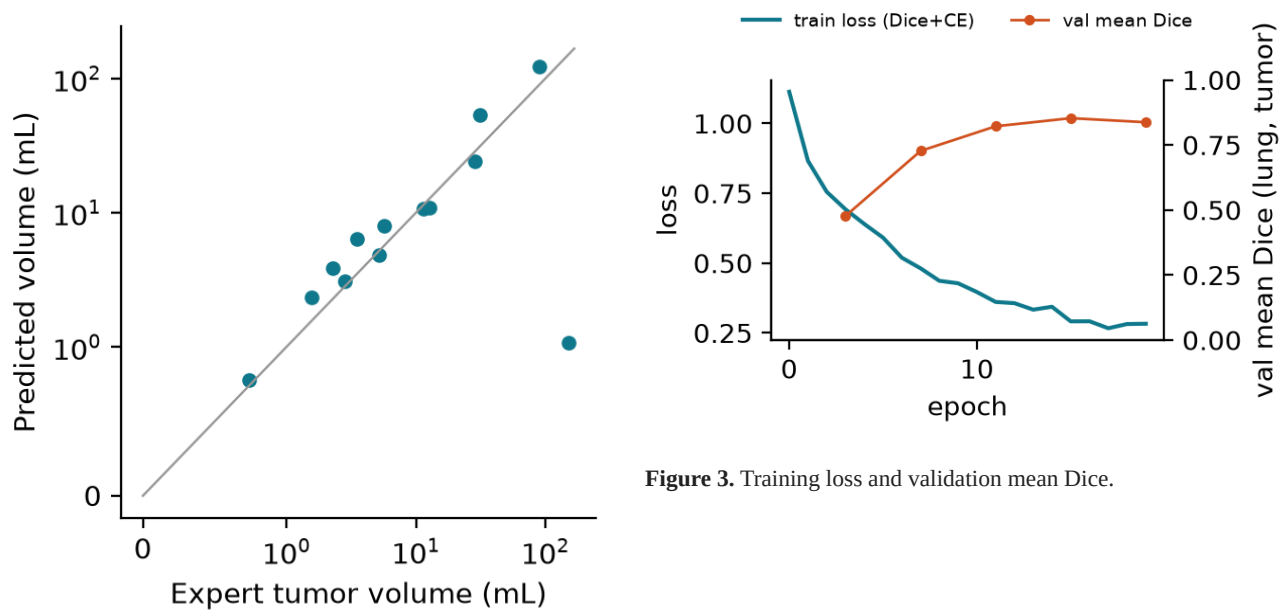


Figure 3. Training loss and validation mean Dice.

Figure 2. Predicted (default post-processing) vs. expert tumor volume, symmetric log axes.

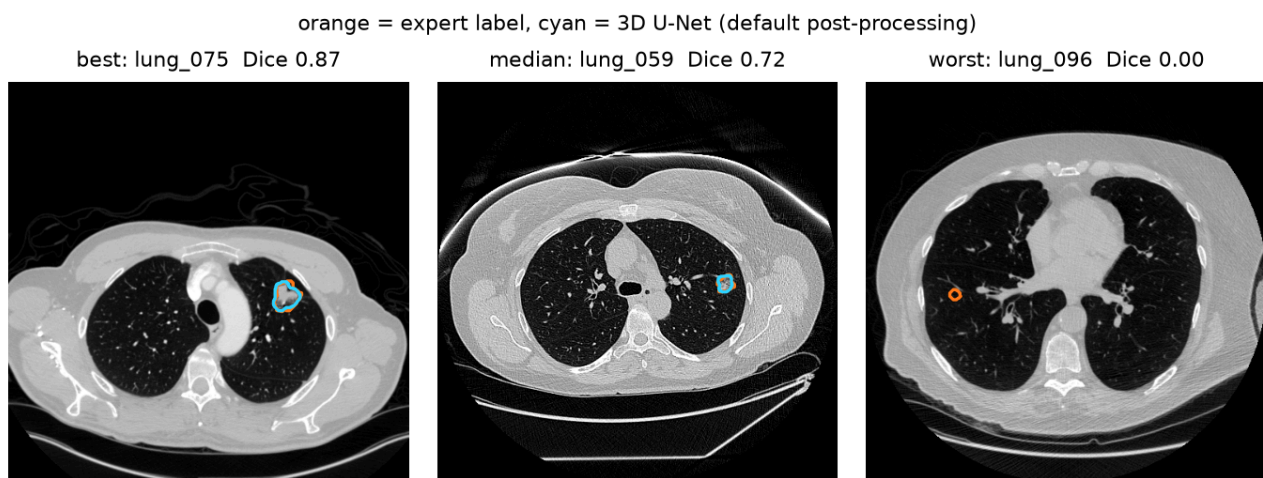


Figure 4. Best, median and worst test patients (axial slice with the largest expert-tumor area). Orange: expert; cyan: 3D U-Net (default post-processing).

Table 3. Per-patient results (test split), sorted by Dice (default rule). Volumes in mL; HD95 in mm (∞ when nothing was predicted).

Case	GT vol.	default post-processing					largest component			Ceiling
		Pred. vol.	Dice	HD95	Lesions found	FP	Dice	HD95	FP	
lung_075	11.4	10.7	0.867	4.1	1/2	1	0.878	2.7	0	0.940
lung_055	28.6	24.0	0.782	9.9	1/1	2	0.760	18.6	0	0.948
lung_078	5.7	8.0	0.772	2.6	1/1	1	0.778	2.5	0	0.958
lung_009	89.5	121.9	0.745	22.8	1/1	1	0.764	9.8	0	0.976
lung_014	31.1	53.2	0.737	174.5	1/1	3	0.852	3.3	0	0.968
lung_044	12.6	10.9	0.728	166.7	1/1	3	0.775	5.3	0	0.942
lung_059	1.6	2.3	0.722	33.7	1/1	1	0.803	1.6	0	0.907
lung_043	2.8	3.1	0.715	51.3	1/1	2	0.763	2.9	0	0.932
lung_029	3.5	6.3	0.680	3.9	1/1	0	0.680	3.9	0	0.929
lung_093	2.3	3.9	0.611	187.5	1/1	2	0.811	2.5	0	0.924
lung_038	5.2	4.8	0.570	204.1	1/1	5	0.679	5.2	0	0.966
lung_028	151.3	1.1	0.009	73.2	0/1	3	0.009	72.7	0	0.974
lung_096	0.7	0.8	0.000	66.0	0/1	1	0.000	66.0	1	0.930

lung_028: expert tumor 151 mL (orange), predicted tumor (cyan), lung label (blue). Inside the expert tumor the network predicted 76% background, 23% lung, 1% tumor.

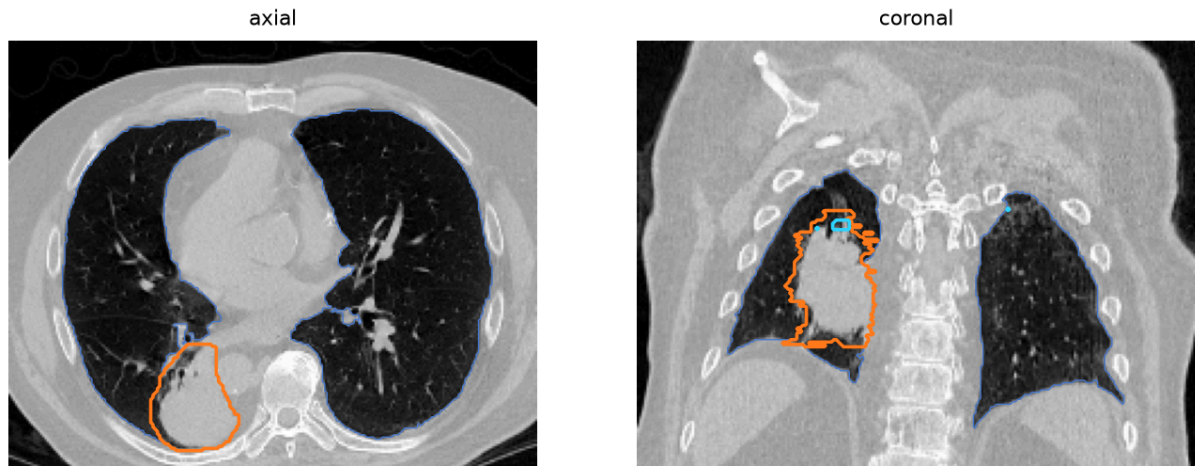


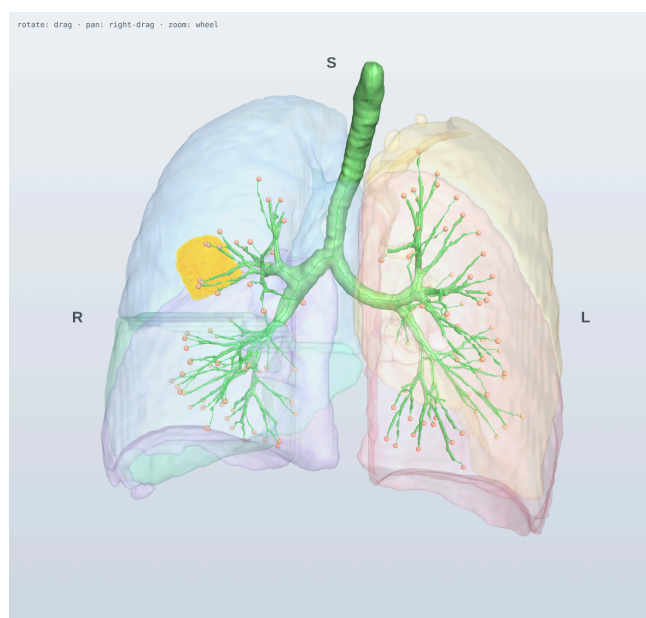
Figure 6. Failure analysis of the largest test tumor on the 1.5 mm grid. Orange: expert tumor; cyan: predicted tumor; blue: threshold-derived lung label.

4.1 Anatomy-level reconstruction

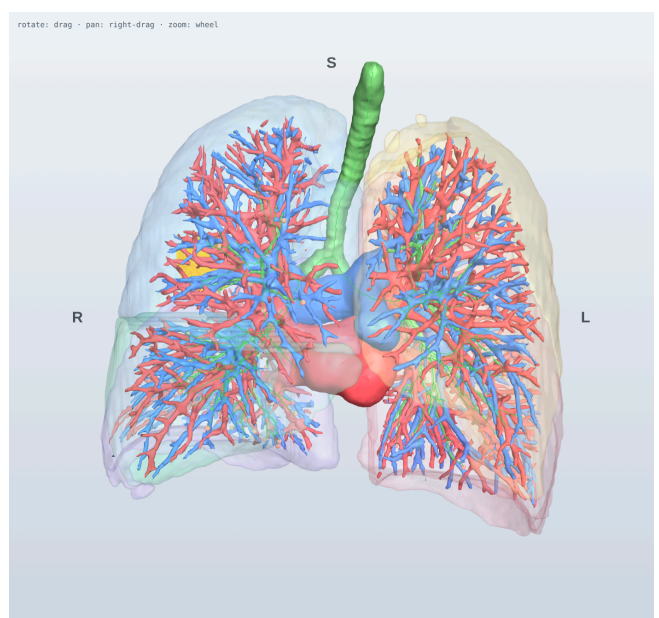
lung_075 has a lobe-derived lung volume of 9.85 L, above the usual adult range; our independent air-threshold estimate agrees, so it reflects the image as stored, but we cannot rule out a slice-spacing error in the file header. For lung_075 we ran only the fast lobe/trachea model, not the airway/vessel model, so it has no vessels and only a coarse trachea. Figure 5 shows reconstructions from real CT. Lobes, airway tree with centreline and endpoints, arteries, veins and tumor are all in patient RAS coordinates, so they can be loaded straight into 3D Slicer or the web viewer. Table 4 summarises the derived anatomy.

Table 4. Anatomy reconstructions. Lung volume in L; airway, artery and vein volumes in mL; airway graph from the pruned centreline (spurs < 5 mm removed).

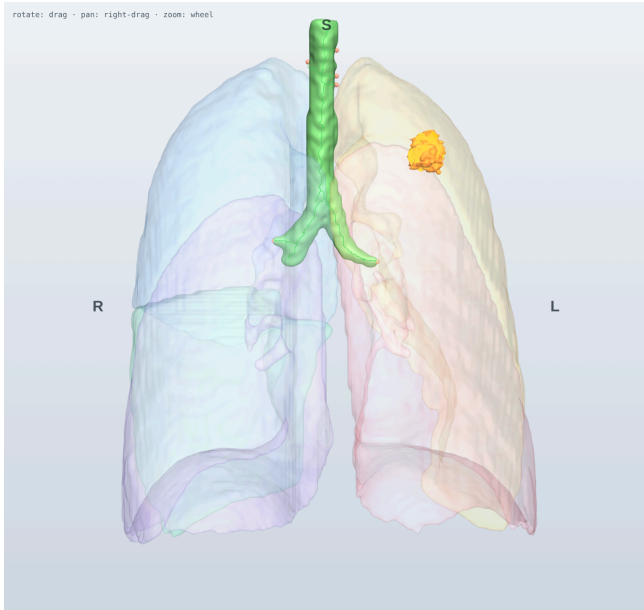
Case	Lung	Lobes	Airway	Branch pts	Endpoints	Max gen.	Arteries	Veins	Tumor lobe	U-Net Dice
lung_014	7.22	5	74	169	125	16	287	281	upper lobe right	0.852
lung_075	9.85	5	78	13	8	12	–	–	upper lobe left	0.878



lung_014: lobes, airway tree with centreline and endpoints, tumor



lung_014: pulmonary arteries (blue) and veins (red) added



lung_075: lobes, airway tree with centreline and endpoints, tumor

Figure 5. Anatomy-level reconstructions of held-out test patients (anterior view; patient right on the viewer's left). Lobes are translucent, airway green with centreline and red endpoints, tumor orange (expert) and yellow wireframe (3D U-Net, largest-component rule).

5 Discussion

What the numbers mean. nnU-Net reports roughly 0.7 Dice on this task [2], but on the official MSD test set (hidden labels). Ours is a 13-patient internal split of the labelled data, so the two cannot be ranked against each other. Our model has about 3M parameters and trained for 20 short CPU epochs; nnU-Net's default is 1000 GPU epochs with a five-fold ensemble. We do not claim state-of-the-art accuracy. The value of this work is a transparent, bounded baseline. The controls show the score is real (shuffled labels near 0), non-trivial (above the threshold rule) and limited partly by our own resampling (the ceiling is below 1 for small tumors).

Failure modes. Across the 13 test patients, Dice showed no clear association with log tumor volume (Pearson $r = 0.03$). The resampling ceiling tends to be lower for smaller tumors ($r = 0.81$ with log volume; Table 3), so our 1.5 mm grid costs more there. The weakest test case (lung_096, expert volume 0.7 mL) reached Dice 0.00. With the default post-processing the model produced 25 false-positive components across 13 scans; the validation-selected largest-component rule reduced this to 1. Lesion sensitivity was unchanged (0.79) on this test set, but the rule would drop any second lesion in a multi-lesion scan. The largest test tumor (lung_028, 151 mL) was essentially missed. Inside it the raw network output was 76% background, 23% lung and 1% tumor. On visual inspection (Fig. 6) it is a posterior mass against the chest wall and spine, with soft-tissue density continuous with the tissue outside the lung. Our lung class is defined by thresholding aerated lung, so during training such tissue looks like the background class. Deriving the lung class from anatomical lung masks, which include the tumor bed, would address this directly; we have not tested it. Other failures were not audited individually.

Limitations. (1) A single split with 13 test patients; the confidence intervals are wide, and cross-validation would be better. (2) No external test set: MSD comes from one institution. (3) Anatomy labels come from TotalSegmentator and are not checked against experts here. The airway generation count comes from a pruned skeleton and is an approximation. Values above about 10 (Table 4 reaches 16) exceed what CT normally resolves and probably reflect skeleton loops, not real branching. (4) The lung class is threshold-derived, not expert-drawn. (5) The system segments tumors that are already known to be present. It does not tell malignant from benign nodules and must not be used for diagnosis or screening. (6) Chest X-ray is supported only for lung-field segmentation in the codebase; this report does not evaluate it.

Future work. An anatomical lung class (see failure analysis), five-fold cross-validation, longer GPU training, external validation on LIDC-IDRI/NSCLC-Radiomics, and expert review of the anatomy meshes.

6 Conclusion

We present a small, CPU-trainable 3D U-Net and an anatomy-level reconstruction pipeline for lung tumor CT, evaluated with a protocol designed to reveal false success. The code, tests, viewer and every number in this report are generated from the public repository.

Reproducibility

```
python scripts/download_datasets.py msd
python -m lungseg.prep_ms3d
python -m lungseg.train --data msd3d --root data/3d --dim 3 --isotropic --patch 64 96 96 \
    --base 12 --batch 2 --epochs 20 --samples-per-volume 8 \
    --tumor-oversample 0.66 --lr 0.003 --n-train 45 --n-val 5 --n-test 13 \
    --val-every 4 --seed 42 --out runs/msd3d
python scripts/evaluate_ms3d.py --ckpt runs/msd3d/best.pt
python scripts/reconstruct_case.py <case> --ckpt runs/msd3d/best.pt
python scripts/build_paper.py --run-tests
```

Test split: lung_044, lung_055, lung_029, lung_038, lung_014, lung_028, lung_043, lung_078, lung_096, lung_059, lung_075, lung_009, lung_093.

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Appendix A Synthetic phantom

During development we used a procedural chest-CT phantom (body, fat, lungs with vessel-like texture, spine, 0–3 lobulated nodules, Gaussian noise) to test the code offline. On this phantom the 2D model reached lung Dice 0.999 and tumor Dice 0.939. These numbers only show that the code works; the phantom is far easier than real CT and we do not use it to support any claim.